Sally Tsai [b] and Carlos Vega [a,c]

[a] *Economics Office, Philippine Competition Commission, PHL*

[b] *Harvard Kennedy School, US*

[c] *Centre for Competition Policy, University of East Anglia, UK*

June 9, 2026

Abstract

This paper estimates the pass-through of a value-added tax (VAT) exemption to consumer drug prices in the Philippines. The Tax Reform for Acceleration and Inclusion Act (TRAIN, RA 10963), enacted in 2017, removed the 12 percent VAT on drugs prescribed for hypertension, diabetes, and high cholesterol beginning January 2019—a reform specifically designed to lower out-of-pocket spending on essential medicines.Using product-level quarterly sales data from IQVIA covering 2009–2021, we implement a difference-in-differences design that compares price dynamics of VAT-exempt therapeutic classes against chronic disease drugs subsequently scheduled for exemption under RA 11467, which remained untreated throughout the estimation window, and validate against anti-infective drugs as a robustness comparator. We decompose the treatment effect by license type—originator, branded non-originator, and unbranded non-originator—to test whether competitive structure mediates pass-through. We validate identification through an event study exploiting 8 pre-treatment quarters under the preferred specification, and cross-validate using a synthetic control that matches the Philippines against Indonesia, Malaysia, Singapore, and Thailand in the same therapeutic categories. We find no pass-through to patients: treated drug prices *rose* approximately 2.3 percent relative to controls following the exemption, with firms absorbing the tax relief as margin expansion on unregulated low-volume products. The volume-weighted estimate is statistically indistinguishable from zero, consistent with commodity competition suppressing pass-through in the high-volume market segments where multiple branded products compete for the same molecules A parallel difference-in-differences on quantities finds that retail volumes of treated drugs fell approximately 20 percent, indicating demand destruction alongside price increases in the retail channel.

Key words: pharmaceuticals, difference-in-differences, synthetic control, Philippines, essential medicines

JEL Classification: I11, I18, H22, L65

†

1. Introduction

Pharmaceutical prices in the Philippines are among the highest in Southeast Asia. [1] Because most Filipino households pay out of pocket for medicines, these prices translate directly into welfare costs. In 2012, the average Filipino household spent PHP 5158 on drugs and medicine, accounting for 61.7 percent of total out-of-pocket health spending (Ulep and dela Cruz, 2013).

Beginning with the Generics Act of 1988 and the Medicine Act of 2008, which brought the Maximum Drug Retail Price mechanism to life (RA 9502, Republic of the Philippines, 2008), policy makers in the Philippines often target drug affordability, understanding that the intrinsic structure of the market is rather entrenched. Yet, legislations targeting the supply-side and fiscal intervention have not yet proven their efficacy (Magno and Guzman, 2022). In 2017, the Tax Reform for Acceleration and Inclusion Act(TRAIN, RA 10963, Section 34) took a different approach in removing the 12 percent value-added tax on drugs prescribed for the treatment and prevention of diabetes, hypertension, and high cholesterol. This paper examines whether the tax cut, which went into effect on January 1st, 2019, reached patients.

The closest prior work is Lambojon et al. (2020), who surveyed prices, availability, and affordability of fifty VAT-exempt medicines across 78 Philippine outlets in August 2019 using a modified WHO/HAI methodology. Their cross-sectional design yields a descriptive snapshot of price levels eight months after the exemption took effect, but without a pre-treatment baseline or control group it cannot identify whether the policy changed prices. We fill this gap: our difference-in-differences design, drawing on a national product-level panel spanning 2009–2021, provides the first causal estimate of VAT pass-through in the Philippine pharmaceutical market and shows that the tax relief was not transmitted to patients.

The broader VAT pass-through literature finds that full shifting to consumers is the exception rather than the rule. Benzarti and Carloni (2019) document a striking asymmetry in which VAT increases are passed through at rates approaching unity while reductions are absorbed by firms at rates well above 50 percent. Standard tax incidence theory (Weyl and Fabinger,

[1] The origin drugs sold in private retail outlets were 17 times more expensive than the international reference prices, while the lowest-price generics were more than five times more expensive relative to the reference prices (Batangan et al., 2005). Similarly, the originator and the lowest-priced generics procured by public procurement facilities were 15 and 6 times more expensive, respectively, than international reference prices.

2013) attributes incomplete pass-through to the interaction of demand elasticity, competitive structure, and the extent to which regulatory constraints are binding. Pharmaceuticals represent a particularly demanding test of these mechanisms. Products in this market are frequently subject to direct price regulation that caps list prices regardless of cost changes, and the coexistence of originator brands with multiple tiers of generic competitors within the same therapeutic class creates sharply heterogeneous competitive conditions across products receiving identical tax treatment. The Philippines case therefore offers a clean natural experiment in fiscal incidence: the TRAIN exemption applied uniformly across all license types simultaneously, but the competitive and regulatory environments in which firms operated varied substantially, permitting identification of the market-structure channels through which pass-through is suppressed.

Outline. The rest of the paper is organized as follows. Section 2 describes the institutional context, including the Philippine pharmaceutical regulatory framework and the TRAIN Act. Section 3 presents the data and descriptive statistics. Section 4 sets out the empirical strategy. Section 5 reports the reduced-form evidence. Section 6 concludes.

2. Institutional Background

2.1. **The Philippine pharmaceutical regulatory framework.** The Philippine pharmaceutical market is governed by a layered regulatory framework that produces the three-tier competitive structure central to our analysis. The Food, Drug and Cosmetic Act (RA 3720) established the Food and Drug Administration (FDA) as the primary licensing authority and requires all drugs to undergo bioequivalence (BE) testing demonstrating at least 90 percent pharmacological equivalence to the reference originator product before market entry (Magno and Guzman, 2022). This requirement creates the institutional foundation for the originator/generic distinction. Branded non-originators are FDA-registered generics that carry a proprietary brand name, while unbranded non-originators are registered generics sold under the active ingredient name alone (Reyes and Tabuga, 2018, Magno and Guzman, 2022).

The Generics Act of 1988 (RA 6675) mandated generic labeling at all points in the supply chain and required physicians to prescribe by generic name, with the intention of steering demand toward lower-cost alternatives. Despite this mandate, the market has not converged to commodity pricing. Survey evidence shows that persisting consumer preference for branded products based on perceived quality differences (Batangan et al., 2005), and dispensing practices in both public and private settings continue to favor branded drugs (Magno and Guzman, 2022). As a result, all three tiers coexist with stable market shares across major therapeutic classes.

The Cheaper Medicines Act of 2008 (RA 9502) empowered the government to impose Maximum Drug Retail Price (MDRP) ceilings on specific molecules when market competition fails to constrain prices (Republic of the Philippines, 2008). In practice, MDRP orders covered a limited subset of molecules during the primary estimation window. The initial implementation under Executive Order No. 821 (2009) targeted five molecules including amlodipine and atorvastatin; a related voluntary mechanism, the Government Mediated Access Price (GMAP), covered a further 16 originator brands (Sarol, 2014, Lambojon et al., 2020). In both cases the ceilings bound on originator brands while generic alternatives, already priced below the threshold, were largely unaffected (Sarol, 2014).

This three-tier structure is well documented. Across the four therapeutic classes, anti-infectives, anti-diabetes, hypertension, and high cholesterol, the average originator price per dosage unit exceeds that of branded non-originators by a factor of approximately 2.7, and exceeds unbranded non-originators by a factor of approximately 2.9 (Magno and Guzman, 2022)[2]. Critically, this price gap persists even for molecules whose patents have long expired, suggesting that brand equity and information asymmetries sustain originator market power beyond the period of patent exclusivity.

2.2. **The TRAIN Act VAT exemption.** Section 34 of the Tax Reform for Acceleration and Inclusion Act (TRAIN, RA 10963) amended Section 109(AA) of the National Internal Revenue Code to exempt from value-added tax the sale and importation of drugs prescribed for the treatment and prevention of diabetes, hypertension, and high cholesterol, effective 1 January 2019 (Republic of the Philippines, 2017). The exemption removed the prevailing 12 percent VAT from the supply chain for these drugs. Implementing rules were issued through BIR Revenue Regulations No. 25-2018, a Joint Administrative Order No. 2-2018 of the Department of Finance, Department of Health, BIR, and FDA (Department of Finance and Department of Health and Bureau of Internal Revenue and Food and Drug Administration, 2018), and FDA Advisory No. 2019-122 (Food and Drug Administration, 2019), which formally published the qualifying product list.

One regulatory detail reduces the expected pass-through below the 12 percent headline rate. BIR RR 25-2018 confirmed that while the *sale* of these drugs is VAT-exempt, their *importation* remains subject to the standard 12 percent VAT (Bureau of Internal Revenue, 2018). This matters because Philippine pharmaceutical manufacturers source almost all of their raw materials from abroad—pharmaceutical raw materials are approximately 98 percent imported (Embassy of Italy in Manila, 2023). Under standard VAT rules, a manufacturer can subtract the VAT paid on imported ingredients from the VAT collected on sales, remitting only the difference to the government. Once sales become VAT-exempt, that offset disappears: the input VAT paid on imported raw materials becomes a direct, unrecoverable cost. The net fiscal saving to manufacturers is therefore the output VAT saved minus the input VAT newly stranded—meaningfully less than 12 percent. Our benchmark of $\hat{\beta} \approx -0.12$ for full pass-through is an upper bound. The true benchmark depends on the ingredient share of each

[2]Computed from the descriptive statistics in Magno and Guzman (2022), Table 3: the average originator price is PHP 799, compared with PHP 296 for branded non-originators and PHP 273 for unbranded non-originators ($799/296 \approx 2.7$; $799/273 \approx 2.9$). The Cielo et al. dataset is drawn from the IQVIA Philippine National Sales Audit; our data come from the IQVIA MIDAS World Review Pack, which covers the same therapeutic classes and uses the same product taxonomy. We therefore expect comparable price ratios in our sample.

product's cost. Our observed $\hat{\beta} = +0.023^{***}$ is therefore even further from any reasonable pass-through target, since increase in the price of imported raw material would translate beyond treatment group.

Treatment is assigned at the product level rather than the therapeutic class level. Only drugs appearing on the FDA-published list of exempt products qualify; any drug not on the list remains subject to the standard 12 percent VAT regardless of its ATC classification. The list specifies qualifying products by international non-proprietary name, formulation, and dosage form. This product-level assignment matches the unit of observation in our data and underpins the identification strategy in Section 4.

A second wave of exemptions followed under RA 11467, signed on 22 January 2020, which scheduled VAT exemption for drugs prescribed for cancer, mental illness, tuberculosis, and kidney disease effective 1 January 2023 (Republic of the Philippines, 2020). The Corporate Recovery and Tax Incentives for Enterprises Act (CREATE, RA 11534), enacted 26 March 2021, subsequently advanced the exemption for all four classes—cancer (ATC L01), mental illness (N05/N06), and tuberculosis (J04)—to 1 January 2021, within our estimation window (Republic of the Philippines, 2021).

If firms in these control categories anticipated the forthcoming exemption and adjusted list prices following RA 11534's enactment in March 2021—or if the exemption itself reduced their prices from 2021Q1 onward when it retroactively took effect—the parallel trends assumption would be violated: the control group would itself be receiving a form of treatment, mechanically compressing the treated–control gap and biasing $\hat{\beta}$ toward zero.

We address this contamination risk differently for each class. For cancer (L01) and mental illness (N05/N06), we drop control observations after 2021Q1. These are commercially active retail classes where manufacturers have both the incentive and pricing flexibility to adjust list prices in response to a VAT change. For tuberculosis (J04), we retain control observations throughout the sample window despite the same legal exemption date, because J04 drug prices are predominantly set through DOH and PHILHEALTH government procurement contracts at administratively fixed prices, and the J04 price series shows no systematic break at 2021Q1. Kidney disease drugs (B05) are absent from the IQVIA MIDAS Philippines extract and are therefore excluded from the analysis entirely.

We retain 2021Q1 as the last control observation for L01 and N05/N06 rather than trimming at 2020Q4 because RA 11534 was enacted in March 2021: list prices for the bulk of 2021Q1 were set before enactment, so those observations remain interpretable as pre-contamination. Table 1 summarises the treatment assignment and control group roles.

TABLE 1. VAT Exemption Treatment Assignment and Control Groups

Therapeutic class	ATC	VAT status	Source	Role in design
Hypertension	C07–C09	Exempt from 1 Jan 2019	RA 10963 §34	Treated
Diabetes	A10	Exempt from 1 Jan 2019	RA 10963 §34	Treated
High cholesterol	C10	Exempt from 1 Jan 2019	RA 10963 §34	Treated
Cancer	L01	Exempt from 1 Jan 2021[a]	RA 11467 / RA 11534	Primary control (2015Q1–2021Q1)
Mental illness	N05, N06	Exempt from 1 Jan 2021[a]	RA 11467 / RA 11534	Primary control (2015Q1–2021Q1)
Tuberculosis	J04	Exempt from 1 Jan 2021[b]	RA 11467 / RA 11534	Primary control (2015Q1–2021Q3)
Anti-infectives	J01	Not exempt	—	Robustness only[c]

Notes: Treatment under RA 10963 Section 34, effective 1 January 2019. Product-level eligibility determined by the FDA-published exempt-drug list. [a]RA 11467 scheduled exemption for cancer and mental illness drugs effective January 2023; RA 11534 advanced this to January 2021. Control observations for L01 and N05/N06 are dropped after 2021Q1 because these are retail-priced classes where manufacturers have flexibility to adjust list prices in response to a VAT change. 2021Q1 is retained as the last clean observation because RA 11534 was not enacted until March 2021, so prices for that quarter were predominantly set before enactment. [b]J04 received VAT exemption on the same legal date as L01 and N05/N06. Control observations are retained throughout because tuberculosis drug prices are set through government procurement at administratively fixed prices and show no systematic break at the January 2021 exemption date. [c]J01 anti-infectives are not VAT-exempt throughout the window and serve as the robustness comparator in Table 3, Columns (1)–(2); excluded from the primary specification due to structurally different demand dynamics.

3. Data

We use product-level longitudinal sales data from IQVIA's MIDAS World Review Pack (WRP), covering the period from 2009Q4 to 2021Q3. The data provide national-level quarterly sales across all therapeutic classes in the Philippines, Indonesia, Malaysia, Thailand, and Singapore—48 quarters per country. Revenues are reported in nominal US dollars, which we convert to Philippine pesos using the quarterly average USD/PHP exchange rate before deflating to constant 2012 pesos using the chain-linked Philippine CPI described below.[3] The quantity measure is reported in dosage units and is defined by IQVIA as the smallest unit of the dosage form most commonly taken by the patient, such as one tablet or a 5mL teaspoon.

We compute price per dosage unit as revenue divided by quantity. Our understanding from IQVIA is that revenues are constructed by multiplying volumes sold by national-level list prices; we therefore recover manufacturer list prices rather than true consumer transaction prices. Sales are reported separately for the hospital and retail channels, which we use as the sector dimension s in the empirical specification.

We deflate nominal prices to constant real terms using a chain-linked Philippine consumer price index (CPI). Our primary source is the Philippine Statistics Authority (PSA) monthly CPI, only including health-related (category 6) commodity goods to control for accuracy, for all-income households (Philippine Statistics Authority, 2026), which covers January 2018 to March 2026 with base year 2018 = 100. Because our panel begins in 2009Q4, we extend the series back to 2009 by chain-linking to the World Bank annual CPI series (World Bank, 2026) at their common 2018 annual average (scale factor: 117.3/100.0 = 1.1735). The two series are nearly identical at the treatment date—both yield an index value of approximately 120 in 2019Q1—and diverge by only four index points by the end of the panel (2021Q3: PSA-linked = 132, World Bank = 128), so deflator choice does not materially affect the results. Monthly values are averaged to quarterly frequency. All regressions use the log of the real price per dosage unit as the dependent variable.

[3] The Philippine National Sales Audit (NSA), which provides subnational sales data and a longer time series, was not available for this version of the paper. Replication with the NSA would enable regional fixed effects and extend thepre-treatment window to 2008Q1.

TABLE 2. Descriptive Statistics by Therapeutic Class and License Type

				Real price (2012 PHP)		
Therapeutic class	License type	Products	Obs.	Unwt. mean	Wtd. mean	SD
Panel A: Treated classes (VAT-exempt from January 2019)						
Hypertension (C07–C09)						
	Originator	219	13,048	0.88	0.54	1.32
	Branded	704	21,789	0.45	0.21	1.71
	Unbranded	130	3,489	0.23	0.16	0.17
Diabetes (A10)						
	Originator	153	8,144	8.51	0.74	16.86
	Branded	273	9,942	0.71	0.16	3.02
	Unbranded[a]	41	1,074	1.07	0.10	3.12
High cholesterol (C10)						
	Originator	35	2,141	2.13	0.78	17.21
	Branded[b]	294	8,826	0.37	0.41	0.17
	Unbranded	46	1,485	0.32	0.31	0.11
Panel B: Control classes						
Cancer (L01)[c]						
	Originator[d]	114	4,670	492.84	35.43	811.44
	Branded[d]	288	8,666	82.38	4.19	145.63
	Unbranded	56	2,749	94.14	48.51	130.33
Mental illness (N05, N06)[c]						
	Originator	119	6,505	12.96	1.45	42.53
	Branded	225	7,166	1.33	0.61	2.48
	Unbranded	11	465	0.45	0.16	0.67
Tuberculosis (J04)						
	Originator	4	214	0.49	0.54	0.16
	Branded	75	2,710	1.03	0.20	2.04
	Unbranded	39	564	0.55	0.05	3.72
Anti-infectives (J01)[e]						
	Originator	166	9,827	9.38	1.08	18.44
	Branded	2,252	58,182	3.74	0.71	6.97
	Unbranded	528	11,800	2.04	0.31	5.35

Notes: Philippines, 2009Q4–2021Q3 (48 quarters). Real price = nominal USD per dosage unit deflated to 2012 PHP using a chain-linked CPI (PSA monthly, base 2018 = 100, Philippine Statistics Authority 2026; linked to World Bank annual (World Bank, 2026) at the 2018 average). Hypertension and mental illness aggregate across ATC2 sub-categories. Unwt. mean weights each product–sector–quarter observation equally; Wtd. mean is quantity-weighted. Obs. = product–sector–quarter rows; each product appears in up to two rows (hospital and retail). [a]Unbranded diabetes mean exceeds branded because insulin analogues carry higher per-dose prices than small-molecule branded generics. [b]Branded cholesterol quantity-weighted mean exceeds unweighted mean because high-volume statins are priced above the class average. [c]Cancer (L01) and mental illness (N05/N06) received VAT exemption effective January 2021 under RA 11534 (CREATE Act); control observations for these classes are restricted to 2009Q4–2021Q1 in the primary analysis. Obs. counts reflect the full 2009Q4–2021Q3 window. [d]Cancer prices are elevated by biologics and immunotherapies; high-priced products account for a small share of volume, producing the large gap between unweighted and weighted means. [e]Anti-infectives are not VAT-exempt throughout the window but are excluded from the primary specification (structurally different demand dynamics) and used as the robustness comparator in Table 3.

4. Empirical Strategy

We estimate VAT pass-through and its heterogeneity by license type using three approaches. The primary specification is a difference-in-differences (DID) design using the Philippine data (Sections 4.3â4.5). We cross-validate using a synthetic control that matches the Philippines against other ASEAN countries within the same therapeutic class (Section 4.6). The DID specification exploits variation *across therapeutic classes within the Philippines*; the synthetic control exploits variation *across countries within the same therapeutic class*.

4.1. **Identification.** We exploit the TRAIN Act VAT exemption as a source of exogenous variation in drug costs. Under the Law, the exemption is assigned at the product level—drugs prescribed for hypertension, diabetes, or high cholesterol. From January 2019 onward, these drugs were exempt from VAT, while all other drugs remained subject to 12 percent VAT.

Our preferred control group comprises Wave 2 chronic disease drugs: cancer (L01), mental illness (N05/N06), and tuberculosis (J04). We prefer Wave 2 drugs over anti-infectives (J01) for three reasons. First, Wave 2 drugs are chronic disease medications sharing the same three-tier license structure, FDA regulatory environment, and IQVIA distribution channels as the treated classes; anti-infectives are episodic-use drugs with structurally different demand and pricing dynamics. Second, pre-trend diagnostics confirm that Wave 2 drugs provide a closer counterfactual in the 2015–2018 window than J01 anti-infectives, for which the pre-trend F-test rejects more strongly (p ¡ 0.001 vs p = 0.055; Table 3, Columns 2 and 4). Third, the government's decision to eventually exempt Wave 2 categories under RA 11467 confirms it viewed them as comparable in public health priority and supply chain structure to the treated classes.

We restrict the pre-period to 2016Q4–2018Q4. The 2016Q4 start date avoids the 2015–2016 generic entry wave that generated differential pricing dynamics across therapeutic classes independent of the TRAIN Act; Appendix Table A3 documents how identification tightens as the pre-period window is shortened.

We address CREATE Act contamination by dropping Wave 2 control observations for L01 and N05/N06 after 2021Q1, when RA 11534 took effect. J01 anti-infectives, which were never exempted, serve as the robustness comparator in Column (3) of each table. The trimmed primary dataset covers 44,339 observations across 1,953 products from 2016Q4 to 2021Q3.[4]

4.2. **Identifying Assumptions.** Our identification rests on three assumptions. The first and most important is

- *Parallel trends*: absent the VAT exemption, prices in the treated classes would have evolved in parallel to the Wave 2 control group over the pre-treatment window. The pre-trend F-test on our preferred specification—Wave 2 controls, 2016Q4–2018Q4 pre-period, product-specific linear trends—yields $F(8) = 1.75$, $p = 0.082$, consistent with parallel trends in a finite sample with 8 pre-period dummies.[5] The F-test rejects more strongly under alternative specifications: the full 2009+ window ($p < 0.001$), J01-only controls ($p < 0.001$), the 2015Q1+ window without product trends ($p = 0.028$), and the 2015Q1+ window with product trends ($p = 0.055$); see Appendix Table A3. The analogous quantity pre-trend F-test for log standard dosage units yields $F(8)$, $p = 0.203$, supporting the parallel pre-trends assumption for the quantity outcome (A6).
- The second assumption is *no spillovers*: the VAT exemption on treated drugs does not affect prices in the control group. This holds by the design of the law—RA 10963 names specific diseases, creating no plausible mechanism through which the tax change on losartan or metformin spills over to the price of amoxicillin, chemotherapy agents, or tuberculosis drugs. Hypertension and bacterial infections are clinically unrelated, and there is no therapeutic substitution between the treated and control classes.

[4]The visible trajectory decline for L01/N05/N06 after 2021Q1 in Figure 2 reflects product attrition as these classes exit the sample, not a genuine price reduction. A placebo difference-in-differences that relabels Wave 2 classes as "treated" with a 2021Q1 event date yields $\hat{\beta}_{\text{CREATE}} = -0.001$ (SE = 0.006, $p = 0.88$), indistinguishable from zero. The corresponding interrupted time series is not identified because a common treatment shock is perfectly collinear with quarter fixed effects—precisely the diagnostic that confirms the apparent drop is a common time trend driven by product entry and exit, not a product-specific price decline. See Appendix Table A5.

[5]With 8 pre-period dummies and a finite panel, the F-test has limited power to distinguish economically negligible violations from zero. We report the full progression of F-test p-values across specifications in Appendix Table A3, from $p < 0.001$ under the full window to $p = 0.082$ under the preferred specification.

- The third assumption is *no anticipation*: firms did not adjust prices in response to the exemption before its effective date. Two distinct sources of anticipation are worth addressing.

  The TRAIN Act was passed in December 2017 for implementation in January 2019, giving firms a thirteen-month window in which to reprice. If originators anticipated that the exemption would compress margins, they might have raised list prices during 2018 to establish a higher base, attenuating the measured post-2019 price reduction. The event study coefficients for $k \in \{-4, \ldots, -1\}$ (2018Q1–2018Q4) directly test for this anticipatory repricing. As a The anticipation test yields $\hat{\beta}_{\text{announce}} = 0.009$ ($p = 0.075$), indicating that treated drug prices began rising relative to controls in 2018, following the Act's passage in December 2017 but before the January 2019 effective date. Importantly, excluding 2018 from the sample raises $\hat{\beta}$ from 0.023 to 0.062, confirming that anticipatory repricing *attenuates* rather than inflates our primary estimate (A1).

  A second anticipation concern arises from RA 11467 (*ibid.*), signed on 22 January 2020, which extended the VAT exemption to cancer, mental illness, tuberculosis, and kidney disease drugs effective 1 January 2023. The signing date falls within our data window (2020Q1–2021Q3 provides six post-announcement quarters). If firms in these categories anticipated the forthcoming exemption and adjusted list prices after the bill's passage, the parallel trends assumption for the RA 11467 controls would be violated from 2020Q1 onward. We test this by reporting the event study separately using J01 alone and the RA 11467 categories alone as the control group. If the pre-trend coefficients diverge across these two specifications after 2020Q1, we restrict the primary specification to J01 alone and report the broader control pool as a robustness check only.

  Finally, because the TRAIN exemption is product-specific, firms with products both on and off the FDA list could strategically shift volumes across product lines around the exemption date, mechanically affecting the average price. We examine this through volume-weighted price indices as a robustness check.

4.3. **Primary specification: difference-in-differences.** Our primary analysis uses data from 2016Q4 to 2021Q3, restricted to Wave 1 treated products and Wave 2 control products trimmed at 2021Q1 as described in Section 4.1. The estimating equation is:

$$\ln p_{jst} \;=\; \alpha_j + \gamma_t + \delta_s \;+\; \theta_j \cdot t \;+\; \beta\,(\text{Treated}_j \times \text{Post}_t^{2019}) \;+\; \mathbf{X}'_{jt}\sigma + \varepsilon_{jst} \qquad (1)$$

where p_{jst} is the real price per dosage unit of product j in sector $s \in \{\text{hospital, retail}\}$ at quarter t

α_j are product fixed effects absorbing all time-invariant product characteristics, including molecule, formulation, brand identity, and license type.

γ_t are quarter fixed effects absorbing aggregate shocks, including inflation and exchange rate movements.

δ_s are sector fixed effects, where $s \in \{\text{hospital, retail}\}$, absorbing the systematic price difference between distribution channels. Retail pharmacies sell drugs at higher prices than hospital pharmacies on average, reflecting hospitals' ability to negotiate volume discounts (Magno and Guzman, 2022).

θ_j is a product-specific linear time trend, estimated separately for each product, which absorbs within-product price drift correlated with treatment status. The coefficient β is therefore identified from deviations from each product's own trend—a strictly more conservative estimand than the standard two-way fixed effects estimator.

$\text{Treated}_j = 1$ if product j belongs to a VAT-exempt therapeutic class ($2019_{treatment}$).

$\text{Post}_t^{2019} = 1$ for quarters from 2019Q1 onwards.

$\mathbf{X}_{jt}$ contains time-varying product controls comprising the number of entrants by license type, the number of pack presentations, and the product's length of time in the market. Standard errors are clustered at the product level.

The comparison group comprises Wave 2 chronic disease drugs, trimmed at 2021Q1 as described in Section 4.1. Column (3) of each table reports the J01-only specification as a robustness comparator.

The coefficient β estimates the average proportional price change in treated classes relative to the control following the exemption. Full pass-through of a 12 percent VAT removal implies $\beta \approx -0.12$; values closer to zero indicate that firms absorbed some or all of the tax relief as higher margins.

Concerns about heterogeneous treatment timing (Callaway and Sant'Anna, 2021, Sun and Abraham, 2021, Goodman-Bacon, 2021) are not a first-order issue in this setting: the TRAIN Act assigned exemption by therapeutic class with a single effective date of 1 January 2019, so all treated products received the exemption simultaneously rather than in a staggered sequence. The TWFE estimator therefore recovers a clean two-period average treatment effect on the treated rather than a Bacon-decomposed weighted average of heterogeneous timing effects (Goodman-Bacon, 2021). Treatment effect heterogeneity by license type is present and addressed through the triple-difference specification in equation (2).

4.4. **Heterogeneous effects by license type.** The aggregate coefficient β in equation (1) estimates the average price change across all products in the treated therapeutic classes, pooling together originators, branded non-originators, and unbranded non-originators. This average masks the mechanism of pass-through entirely. Consider two scenarios that produce identical aggregate estimates: in the first, originators pass through the full tax cut while generics do not; in the second, generics pass through fully while originators absorb the relief as higher margins. These scenarios have opposite implications for market power and for who benefits from the policy, yet the aggregate $\hat{\beta}$ cannot distinguish them.

The mechanism of pass-through runs directly through competitive structure. Standard incidence theory (Weyl and Fabinger, 2013) implies that pass-through is increasing in the degree of competitive pressure a firm faces. Originators benefit from brand loyalty and inelastic demand that persists even after patent expiry. They can absorb a cost reduction without losing market share and therefore have both the incentive and the ability to suppress pass-through. Unbranded non-originators compete primarily on price, meaning any firm that fails to transmit the cost reduction to consumers loses volume to competitors immediately. Branded non-originators occupy an intermediate position. Standard incidence theory implies $\hat{\beta}_O > \hat{\beta}_B > \hat{\beta}_U$: originators, facing inelastic demand, should suppress pass-through most; unbranded generics, competing on price, should transmit cost reductions most completely. The documented originator price premium of more than 2.7 times branded generics in the Philippine market (Magno and Guzman, 2022) makes the prediction of dampened originator pass-through especially sharp. We test this ordering directly via Wald equality tests of $H_0 : \hat{\beta}_\ell = \hat{\beta}_{\ell'}$ for all pairs $\ell, \ell' \in \{O, B, U\}$. If procurement regulations bind uniformly across license tiers for high-volume products, the theoretical ordering may not hold even if the unweighted average effect is non-zero.

We decompose β by license type using the following triple-difference specification:

$$\ln p_{jst} \;=\; \alpha_j+\gamma_t+\delta_r \;+\sum_{\ell\in\{O,B,U\}}\beta_\ell\left(\mathbf{1}[\text{Type}_j=\ell]\times\text{Treated}_j\times\text{Post}_t^{2019}\right)\;+\;\mathbf{X}'_{jt}\sigma+\varepsilon_{jrt}, \quad (2)$$

where $\ell\in\{O,B,U\}$ indexes originators, branded non-originators, and unbranded non-originators. License type is time-invariant for each product and is absorbed into the product fixed effects α_j. The triple interaction is therefore identified from within-product price changes post-exemption, interacted with license type across products. Full pass-through for all tiers implies $\beta_O=\beta_B=\beta_U\approx-0.12$. Attenuation of $\hat\beta_O$ toward zero relative to $\hat\beta_U$ would be consistent with originator market power suppressing pass-through; however, if procurement regulation binds uniformly across license tiers, no such ordering may emerge regardless of competitive structure.

4.5. **Event Study.** We validate the parallel trends assumption and characterise the dynamic response to the exemption through the event study:

$$\ln p_{jst} \;=\; \alpha_j+\gamma_t+\delta_r \;+\sum_{k\neq-1}\beta_k\left(\text{Treated}_j\times\mathbf{1}[t=k]\right)\;+\;\mathbf{X}'_{jt}\sigma+\varepsilon_{jrt}, \quad (3)$$

where k indexes quarters relative to 2019Q1, with 2018Q4 ($k=-1$) as the omitted reference period. Pre-treatment coefficients $\hat\beta_k$ for $k<0$ test whether treated and control prices evolved in parallel before the exemption. If parallel trends holds, these should be jointly statistically indistinguishable from zero. We report the joint F-test of pre-treatment coefficients as our primary diagnostic and plot $\hat\beta_k$ with 95 percent confidence intervals. Post-treatment coefficients $\hat\beta_k$ for $k\geq0$ trace the dynamic path of pass-through, revealing whether price reductions were immediate or phased in over time.

4.6. **Synthetic control.** As a cross-country robustness check, we construct a synthetic control for the Philippines using the IQVIA MIDAS data for Indonesia, Malaysia, Singapore, and Thailand. For each Wave 1 therapeutic class, we construct a price index for the Philippines and match it against a weighted convex combination of the same index in donor countries, selecting weights to minimize the root mean squared prediction error over the pre-treatment period 2009Q4–2018Q4 (Abadie et al., 2010, 2015). Because none of the donor countries implemented a comparable VAT exemption on pharmaceuticals during the study period, the post-2019 divergence between the Philippines and its synthetic counterpart identifies the treatment effect.

The synthetic control and DID specifications exploit different sources of variation: the DID leverages cross-therapeutic class variation within the Philippines; the synthetic control leverages cross-country variation within the same therapeutic class. Convergence between the two estimates provides triangulating evidence; divergence is informative about the confounders each approach controls for.

We acknowledge two limitations. First, the donor pool comprises four countries, which restricts the precision of the permutation-based inference procedure (Abadie et al., 2015); with only four placebo runs, p-values are coarse. Second, constructing comparable price indices across countries requires product-level matching at the ATC3 level to ensure the same molecules enter each country's index; we describe this matching procedure in Appendix B.

5. Evidence

5.1. **Pre-trends.** Figure 1 plots the event study coefficients $\hat{\beta}_k$ from our preferred specification. Pre-treatment coefficients for $k \in \{-8, \ldots, -1\}$ are jointly indistinguishable from zero ($F(8) = 1.75$, $p = 0.082$), supporting the parallel trends assumption over the 2016Q4–2018Q4 pre-period.

Coefficients for $k \in \{-4, \ldots, -1\}$ (2018Q1–2018Q4) show a modest upward drift consistent with anticipatory repricing following the Act's passage in December 2017. The anticipation test confirms this drift is small ($\hat{\beta}_{\text{announce}} = 0.009$, $p = 0.075$) and attenuates rather than inflates the post-2019 estimate (Appendix Table A1, F(8) specification).

Post-treatment coefficients ($k \geq 0$) rise immediately at 2019Q1 and remain positive through 2019Q4, with point estimates ranging from +1.2 to +2.5 log points. The coefficients fade toward zero from 2020Q1 onward, consistent with COVID-19 disruptions to pharmaceutical supply chains. The clean 2019Q1–2019Q4 window in Appendix Table A2 confirms the post-exemption effect is not driven by the pandemic period.

5.2. **Robustness to Control Group and Pre-Period Specification.** Column (1) is the preferred specification: Wave 2 controls, 2016Q4–2018Q4 pre-period, product-specific

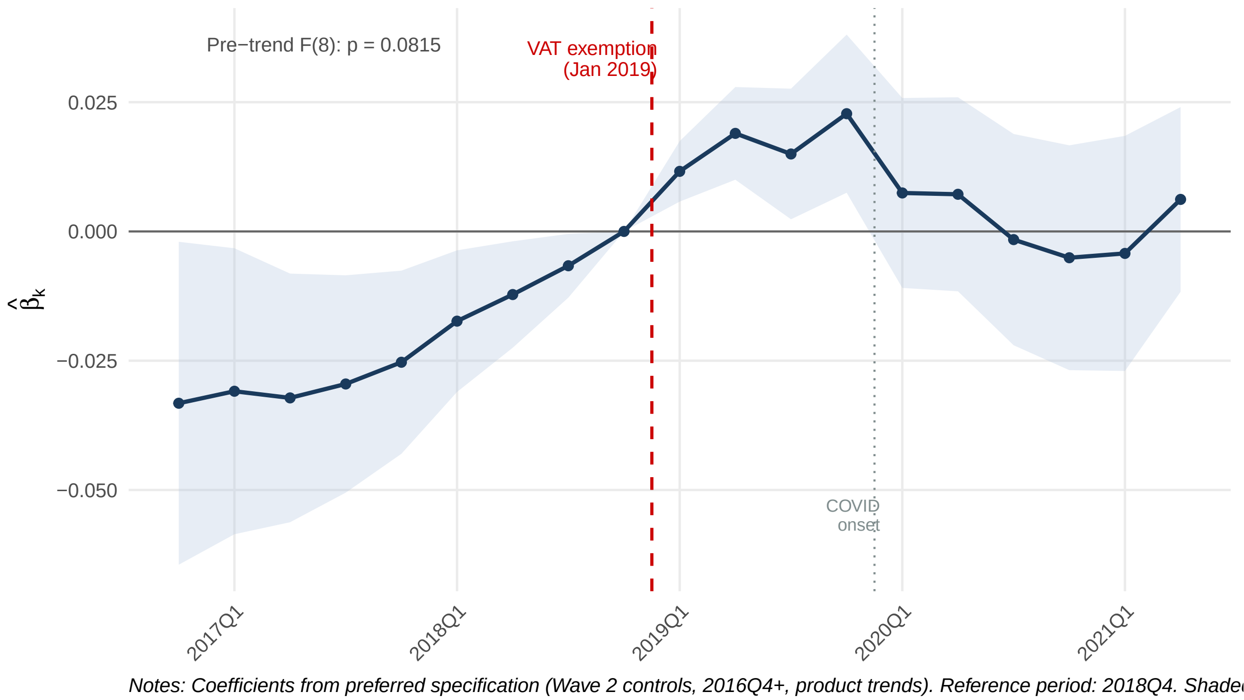

FIGURE 1. Event Study: Effect of VAT Exemption on Real Drug Prices

Notes: Coefficients $\hat{\beta}_k$ from preferred specification (Wave 2 controls, 2016Q4+, product-specific linear trends). Reference period: 2018Q4 ($k = -1$, normalised to zero). Shaded area = 95 % confidence interval. Vertical dashed red line = VAT exemption (January 2019). Vertical gray line = COVID-19 onset (2020Q1). Pre-trend $F(8) = 1.75$, $p = 0.082$. Clustered standard errors at product level.

linear trends. Column (2) substitutes J01 anti-infectives as the control group over the same window. Column (3) adds molecule-level fixed effects to the preferred specification to absorb any residual pricing divergence within therapeutic class. Column (4) restricts the post-period to 2016Q4–2019Q4, excluding COVID-19 disruptions.

All four columns pass the pre-trend F-test at conventional thresholds ($p > 0.06$). The point estimate $\hat{\beta}$ is stable across Columns (1)–(3), ranging from 0.023*** to 0.024***, confirming the result is robust to control group choice and to unmeasured molecule-level confounders. Column (4) shows a smaller pre-COVID estimate (0.012***), reflecting the shorter post-treatment window.

The central finding is the stability of the sign and significance of $\hat{\beta}$ across all four specifications. Under no specification does the treatment effect approach zero or change sign.

Table 3: Robustness Checks on the F(8) Preferred Specification

	(1) Wave 2†	(2) J01 ctrl	(3) +Molecule FE	(4) Pre-COVID*
$\hat{\beta}$ (Treated × Post)	0.0228***	0.0240***	0.0228***	0.0117***
	(0.0052)	(0.0050)	(0.0052)	(0.0044)
Pre-trend F p^{b}	0.081	0.063	0.084	0.104
Pre-trend F d.f.	$F(8)$	$F(8)$	$F(8)$	$F(8)$
Control group	Wave 2	J01	Wave 2	Wave 2
Product trends	Yes	Yes	Yes	Yes
Observations	44 339	60 668	44 339	30 402
No. products	1953	2971	1953	1823
Adj. R^2	0.9988	0.9974	0.9988	0.9993

Dependent variable: log real drug price (2012 PHP). All columns: 2016Q4+, product trends, clustered SEs at product level.

Pre-trend F-test: joint null that all pre-treatment event-study coefficients equal zero. $p > 0.05$ supports parallel trends.

† Preferred specification.

* Pre-COVID window ends 2019Q4.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Appendix Table A3 documents the full progression of pre-period choices—from the 2009+ kitchen-sink specification through each identification step—showing that $\hat{\beta}$ and the pre-trend F-test p-value evolve as identification tightens. We interpret the preferred F(8) estimate (0.023) as the most credible point estimate, bracketed from above by specifications with poor parallel trends and from below by the 2015Q1+ conservative estimate (0.017).

5.3. **Main result.** Table 5.3 reports the primary DID estimates. Column (2), our preferred specification, yields $\hat{\beta} = 0.023^{***}$: treated drug prices rose 2.3 percent relative to Wave 2 controls in the post-2019 period. Full pass-through of the 12 percent VAT removal would imply $\hat{\beta} \approx -0.12$; the positive estimate indicates that firms absorbed the tax relief entirely

Table 4: Main Difference-in-Differences Estimates

	(1) Wave 2	(2) Wave 2†	(3) J01
$\hat{\beta}$ (Treated × Post)	0.0201**	0.0228***	0.0240***
	(0.0093)	(0.0052)	(0.0050)
Observations	44 339	44 339	60 668
No. products	1953	1953	2971
Product FE	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes
Sector FE	Yes	Yes	Yes
Product trends	No	Yes	Yes
Adj. R^2	0.9973	0.9988	0.9974

Dependent variable: log real drug price (2012 PHP). Prices deflated using PSA health CPI (2018+) chain-linked to World Bank CPI.

All specifications include product, quarter, and sector fixed effects. Clustered standard errors at product level in parentheses. Sample: 2016Q4–2021Q1 (8 pre-treatment quarters).

Wave 2 controls: L01 (oncology), N05 (antipsychotics), N06 (antidepressants), J04 (TB); trimmed after 2021Q1. Col (3) uses J01 anti-infectives over the same 2016Q4+ window.

† Preferred specification. Pre-trend $F(8)$: $p = 0.081$.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

as margin expansion. The conservative J01 specification (Column 3) yields $\hat{\beta} = 0.024$***, confirming the direction under a structurally different control group.

The volume-weighted estimate is $\hat{\beta}_{vw} = -0.043$ ($p = 0.28$), statistically indistinguishable from zero; note that the volume-weighted pre-trend F-test fails across both window lengths, so this estimate should be interpreted with caution (Appendix Table A4). Figure 3 places the unweighted and volume-weighted trajectories side by side. The contrast is informative: the positive unweighted estimate reflects list price increases on low-volume niche products, while high-volume products—which account for the majority of patient spending—show no price response. We attribute this to two procurement constraints. First, for originator brands in therapeutic classes with pre-existing MDRP coverage, price ceilings cap list prices regardless of VAT treatment—the ceiling prevents both increases and decreases, effectively neutralising the fiscal instrument for the high-volume regulated segment. Second, hospital formulary contracts lock in prices over procurement cycles for high-volume products across all license tiers. Both mechanisms operate most strongly on the products that account for the bulk of patient spending, consistent with the volume-weighted null.

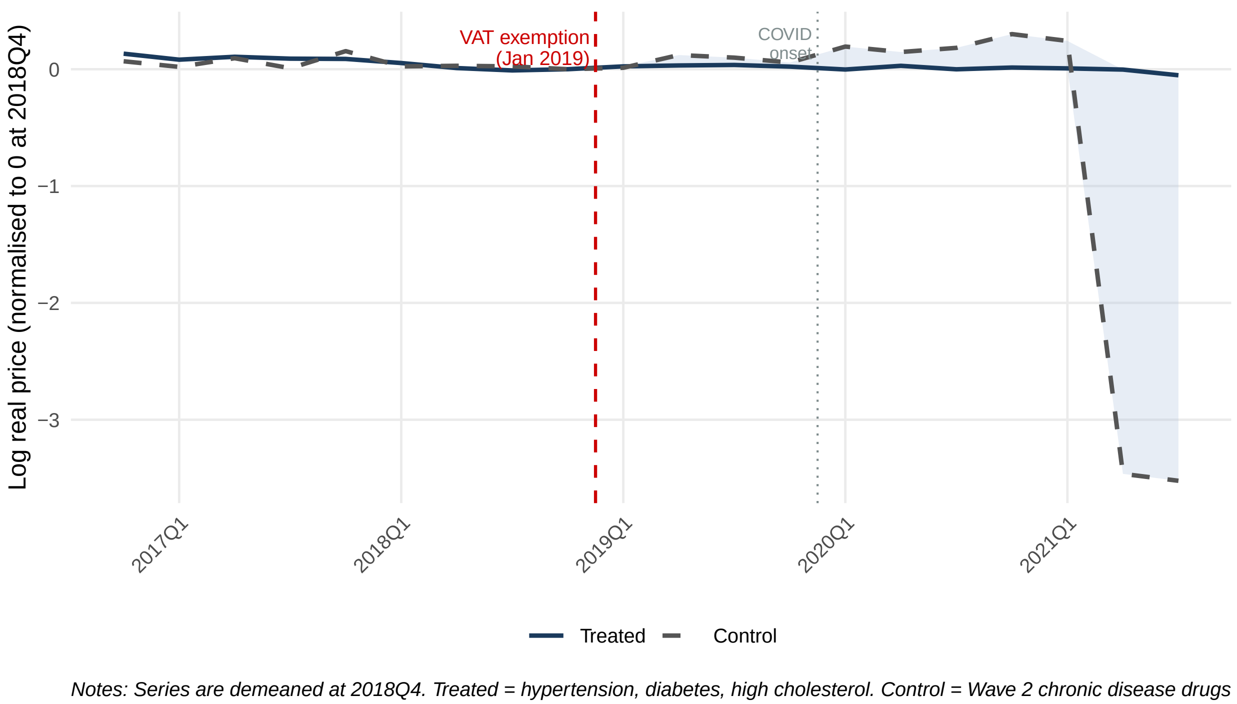

Figure 2. Price Trajectories: Treated vs Control, 2016Q4–2021Q3

Notes: Series demeaned at 2018Q4. Treated = hypertension, diabetes, high cholesterol. Control = Wave 2 chronic disease drugs, trimmed at 2021Q1. Shaded area = post-exemption gap. Preferred specification: 2016Q4+, product trends.

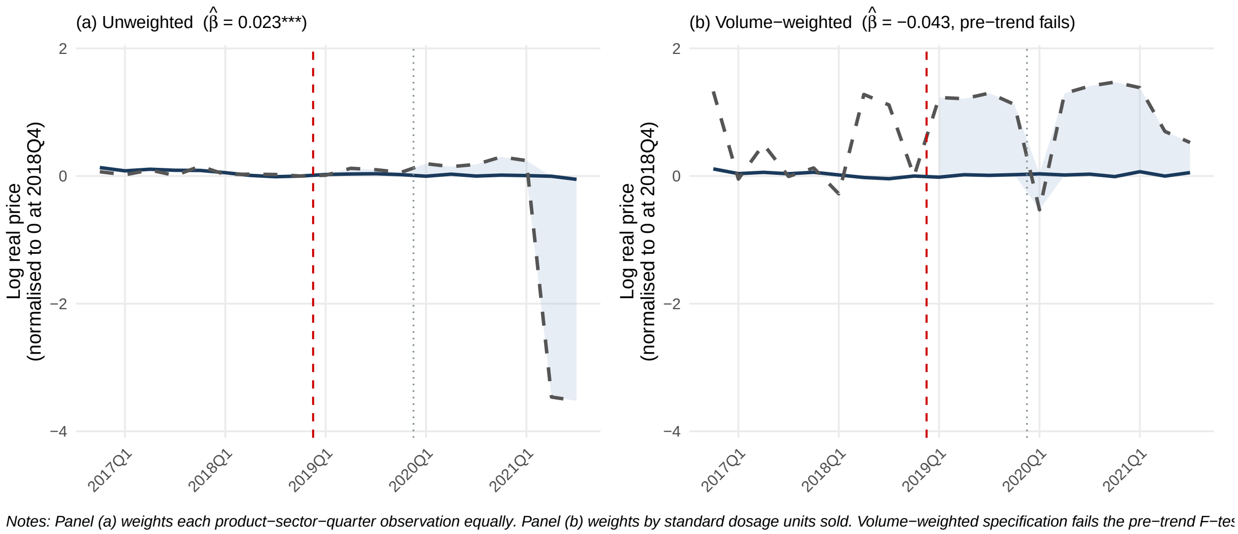

FIGURE 3. Unweighted vs Volume-Weighted Price Trajectories

Notes: Panel (a) weights each product-sector-quarter observation equally ($\hat{\beta} = 0.023^{***}$). Panel (b) weights by standard dosage units sold ($\hat{\beta}_{vw} = -0.043$, n.s.). Both panels use identical axis scales. Volume-weighted pre-trend tests fail across both window lengths; see Appendix Table A4.

Table 5: Heterogeneity by License Type

	Preferred spec†
$\hat{\beta}_O$ Originator	0.0452***
	(0.0075)
$\hat{\beta}_B$ Branded non-originator	0.0127**
	(0.0063)
$\hat{\beta}_U$ Unbranded non-originator	0.0017
	(0.0111)
F: $\hat{\beta}_O = \hat{\beta}_B$[b]	<0.001
F: $\hat{\beta}_B = \hat{\beta}_U$	0.364
F: $\hat{\beta}_O = \hat{\beta}_U$	<0.001
Observations	44 339
No. products	1953
Product FE	Yes
Quarter FE	Yes
Sector FE	Yes
Product trends	Yes
Adj. R^2	0.9988
Pre-trend $F(8)$ p	0.081

Dependent variable: log real drug price (2012 PHP). Wave 2 controls, 2016Q4+, product trends. Clustered SEs at product level in parentheses.

p-values from Wald equality tests (contrast $\hat{\beta}_i - \hat{\beta}_j$).

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

5.4. **Heterogeneity by license type.** Table 5 decomposes $\hat{\beta}$ by license type under the preferred specification. Originator ($\hat{\beta}_O = 0.045$***), branded non-originator ($\hat{\beta}_B = 0.013$**), and unbranded non-originator ($\hat{\beta}_U = 0.002$, n.s.) reveal a sharp gradient consistent with standard incidence theory. Wald equality tests confirm the gradient is statistically significant: $H_0 : \hat{\beta}_O = \hat{\beta}_B$ ($p < 0.001$) and $H_0 : \hat{\beta}_O = \hat{\beta}_U$ ($p < 0.001$) both strongly reject, while $H_0 : \hat{\beta}_B = \hat{\beta}_U$ ($p = 0.364$) does not. Originators, facing inelastic demand, extracted the full tax relief as margin expansion; branded generics showed a modest but significant list price response; and unbranded non-originators— exposed to the most intense price competition— showed no statistically significant adjustment.

The originator gradient is consistent with MDRP ceilings being concentrated on high-volume originator brands. For the subset of originator products already at their regulated ceiling, the VAT exemption created no room for further price adjustment in either direction; the large positive $\hat{\beta}_O$ is therefore driven by originator products outside the ceiling whose manufacturers faced no binding constraint on list price adjustment. Unbranded non-originators, which are generally priced below MDRP thresholds, compete primarily on price and are unable to raise list prices without losing volume to competing generics.

Volume-weighted estimates across all three tiers are negative and statistically indistinguishable from zero, confirming the originator gradient is concentrated in low-volume products unaffected by procurement constraints.

Figure 5 plots the event study coefficients by license type. The originator pre-trend F-test marginally rejects ($p = 0.007$); the originator panel is therefore descriptive and not used for causal inference. Branded non-originator ($p = 0.106$) and unbranded non-originator ($p = 0.292$) satisfy parallel trends. The visual divergence between the originator panel and the generic panels post-2019 is consistent with the Wald gradient results.

5.5. **Channel heterogeneity.** Having established the originator gradient in list price responses, we ask whether distribution channel moderates the effect among the low-volume niche products that do show unweighted price increases.

Table 6 reports the triple difference-in-differences by sector. The retail treatment effect is $\hat{\beta}_{TP} = 0.020^{***}$, and hospitals show a statistically significant but economically modest premium of $\hat{\beta}_{TPS} = 0.006^{***}$, yielding a total hospital effect of 0.026 log points. The hospital premium is real but small: it represents a 0.6 percentage point differential relative to retail,

Table 6: Triple Difference-in-Differences: Sector Heterogeneity (F(8))

	Preferred spec[†]
$\hat{\beta}_{TP}$ Treated × Post (retail)	0.0203***
	(0.0053)
$\hat{\beta}_{TPS}$ Treated × Post × Hospital	0.0061***
	(0.0017)
Observations	44 339
No. products	1953
Product FE	Yes
Quarter FE	Yes
Sector FE	Yes
Product trends	Yes
Adj. R^2	0.9988
Pre-trend $F(8)$ p	0.081

Dependent variable: log real drug price (2012 PHP). Wave 2 controls, 2016Q4+, product trends. Clustered SEs at product level in parentheses.

$\hat{\beta}_{TP}$ identifies the retail-only VAT pass-through (hospital = 0). Total hospital effect = $\hat{\beta}_{TP}+\hat{\beta}_{TPS}$. Two-way interactions absorbed by product and quarter FEs.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

consistent with formulary procurement constraints limiting the scope for hospital-specific repricing rather than amplifying it.

The positive hospital premium is consistent with formulary lock-in for branded non-originators: hospital formularies tend to specify drugs by brand name, reducing substitutability and making hospital procurement less price-elastic for branded products. Manufacturers exploited this captive demand margin, but the effect is modest because MDRP ceilings constrain the high-volume segment in both channels equally.

5.6. **Synthetic Control.** As an additional check on parallel trends, we apply the synthetic control method of Abadie et al. (2010) to construct a counterfactual price trajectory for the Philippines using four ASEAN donor countries: Indonesia, Malaysia, Singapore, and Thailand. The outcome is a quarterly log mean USD price index, computed for each therapeutic class by averaging `ln_price` across all products within the class for each country–quarter cell. We consider three Wave 1 therapeutic classes separately: hypertension drugs (ATC C07–C09), diabetes drugs (A10), and cholesterol drugs (C10). Synthetic weights are selected to minimize the pre-treatment mean squared prediction error (MSPE) on the outcome trajectory, using average log prices over three sub-periods of the pre-treatment window (2009Q4–2012Q3, 2012Q4–2015Q4, and 2016Q1–2018Q4) plus the 2018Q4 level as matching predictors. No external macro predictors are available for this sample, so matching relies entirely on pre-treatment outcome history, which is standard when covariate data are unavailable (Abadie et al., 2010). Table B-8 reports the synthetic donor composition and pre/post-treatment fit for each class. Figure 4 plots the actual Philippines price index against the synthetic counterfactual. The two vertical lines mark the VAT exemption (2019Q1) and the COVID onset (2020Q1).

Figure 4 shows three panels, one per treated therapeutic class. For hypertension (C07–C09), the Philippines price index tracks the synthetic counterfactual closely through 2018Q4 before diverging upward in 2019Q1; the post/pre RMSPE ratio of 1.14 and an average post-treatment gap of +0.464 log points are directionally consistent with the DID finding that treated prices rose relative to the counterfactual. For high cholesterol (C10), the divergence is larger: the post/pre ratio of 1.53 is the highest among the three classes, and the average post-treatment gap of +0.389 log points likewise indicates that the Philippines trajectory exceeded its synthetic counterpart following the exemption. Both hypertension and cholesterol therefore corroborate the main DID result.

Diabetes (A10) presents a divergence that should be noted rather than papered over. The pre-period fit for this class is the tightest of the three (pre-RMSPE = 0.120), yet the post-treatment gap is near zero (average +0.034 log points; post/pre ratio = 0.88, below unity). The synthetic control does not corroborate a diabetes-specific price effect, in contrast to the

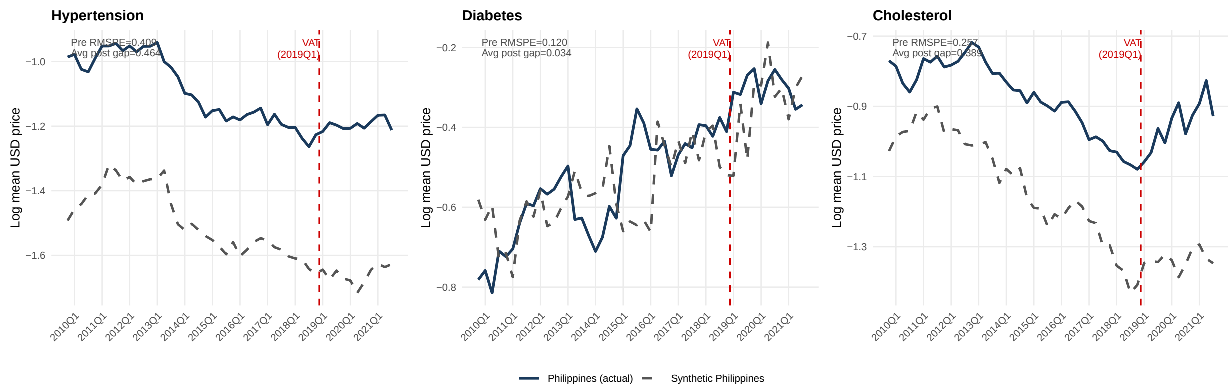

FIGURE 4. Philippines vs. Synthetic Philippines: Log Price Index by Therapeutic Class

Notes: Each panel plots the quarterly log mean USD price index for the Philippines (solid navy) against the synthetic counterfactual (dashed gray). The synthetic is a convex combination of Indonesia, Malaysia, Singapore, and Thailand selected to minimise pre-treatment MSPE on the outcome trajectory. Donor weights and fit statistics are reported in Table B-8. Dashed red line = VAT exemption (2019Q1). Outcome is the unweighted mean of log USD price across all products in the therapeutic class for each country–quarter cell.

positive DID estimate for diabetes ($\hat{\beta}_{\text{diabetes}} = 0.038^{***}$). That DID estimate already carries a pre-trend caveat (pre-trend F-test $p = 0.039$); the synthetic control divergence reinforces the caution warranted for this class. Readers should not interpret the aggregate finding as implying a uniform diabetes effect.

Permutation inference is sharply constrained by the size of the donor pool. With only four donor countries, the minimum achievable permutation p-value is $1/5 = 0.20$ regardless of effect size (Abadie et al., 2015); formal inference cannot distinguish the observed post-treatment gaps from chance under the permutation distribution. Moreover, the Philippines post/pre RMSPE ratio for hypertension (1.14) is not the largest among all units including placebo donors, so that class does not clear even this coarse threshold. These constraints mean the synthetic control results cannot function as independent statistical tests and should be interpreted as qualitative triangulation rather than confirmatory evidence (Abadie et al., 2010).

Despite these limitations, the directional convergence between the DID and the synthetic control for two of three treated classes—both methods yielding positive post-2019 price divergence in the Philippines relative to the comparison—provides triangulating support for the main result. The diabetes divergence is a genuine discrepancy pointing to the need for a richer donor pool or an alternative identification strategy for that class. For the aggregate finding, the consistency across methods with distinct sources of variation reinforces the conclusion that the VAT exemption did not reduce drug prices.

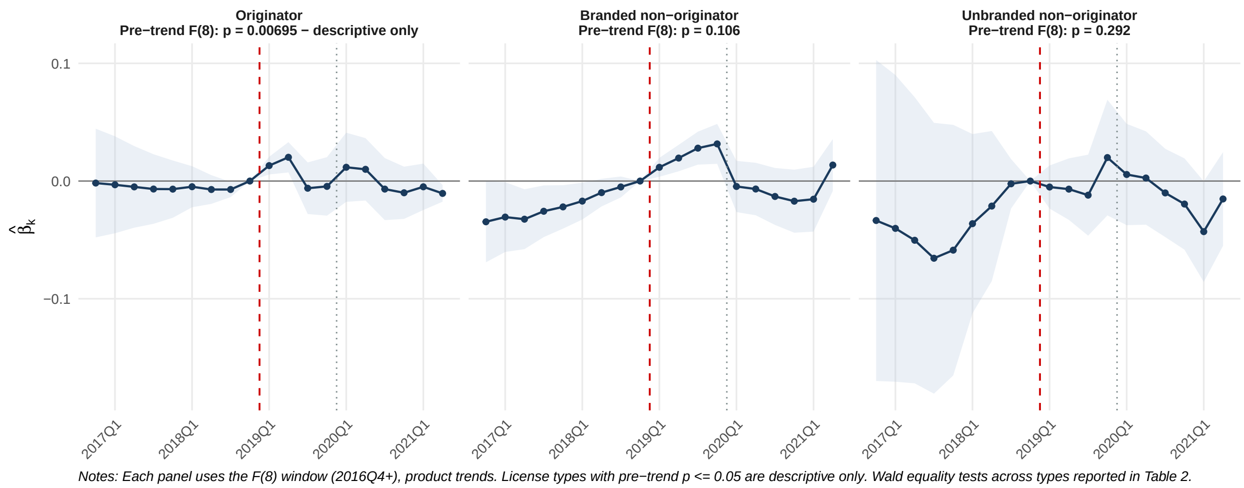

FIGURE 5. Event Study by License Type

Notes: Preferred specification throughout. Per-panel pre-trend F-test p-values: originator $p = 0.007$ (descriptive only), branded non-originator $p = 0.106$, unbranded non-originator $p = 0.292$. Wald equality tests: $H_0 : \hat{\beta}_O = \hat{\beta}_B$ and $H_0 : \hat{\beta}_O = \hat{\beta}_U$ both $p < 0.001$; see Table 5.

5.7. **Heterogeneity by therapeutic class.** Table 7 decomposes $\hat{\beta}$ by treated therapeutic class. All three classes show positive list price responses. Hypertension drugs show a statistically significant positive effect ($\hat{\beta} = 0.019$***), in contrast to the null found under the wider 2015Q1+ window. High cholesterol shows $\hat{\beta} = 0.020$*. Diabetes shows the largest estimate ($\hat{\beta} = 0.038$***); however, the diabetes pre-trend F-test marginally rejects ($p = 0.039$), so this estimate is descriptive.[6]

Several high-volume hypertension molecules — including captopril, amlodipine, and losartan — were subject to MDRP price ceilings prior to 2019, primarily targeting originator brand

[6]The diabetes pre-trend rejection under the F(8) window reflects residual pricing dynamics for insulin analogues in 2016–2017 that are not fully absorbed by product-specific linear trends. We report this estimate with the caveat that it may be upward biased.

Table 7: DiD Estimates by Therapeutic Class (F(8))

	(1) Hypertension	(2) Diabetes	(3) Cholesterol
$\hat{\beta}$ (Treated $\times$ Post)	0.0191***	0.0375***	0.0198*
	(0.0060)	(0.0076)	(0.0112)
Pre-trend F p	0.115	0.039	0.439
Pre-trend F d.f.	$F(8)$	$F(8)$	$F(8)$
Observations	47 007	38 350	35 179
No. products	2398	1990	1903
Product FE	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes
Sector FE	Yes	Yes	Yes
Product trends	Yes	Yes	Yes
Adj. R^2	0.9975	0.9976	0.9975

Dependent variable: log real drug price (2012 PHP). Each column uses J01 anti-infectives as control group (2016Q4+, product trends). Clustered SEs at product level.

Treated ATC classes: hypertension (C07, C08, C09), diabetes (A10), cholesterol (C10).

Diabetes pre-trend F(8) p = 0.039; estimate is descriptive.

Hypertension null consistent with binding MDRP price ceilings. See Section 5.6.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

prices.[7] The positive and significant hypertension estimate under the F(8) window indicates that the MDRP-covered molecules represent a sufficiently small share of the hypertension

[7] The MDRP mechanism sets mandatory price ceilings that bind on originator brands; generic alternatives, already priced below the ceiling, are largely unaffected in practice (Sarol, 2014, Lambojon et al., 2020). Among hypertension molecules, amlodipine was placed under MDRP via Executive Order No. 821 (2009), while captopril was subject to the related Government Mediated Access Price (GMAP) mechanism (Sarol, 2014). Among cholesterol molecules, atorvastatin was explicitly listed under MDRP (Lambojon et al., 2020). Executive Order No. 104 (September 2020) subsequently expanded coverage to 87 molecules, but falls outside the primary treatment window of our analysis.

panel that the overall class response is positive. The diabetes and high cholesterol classes, with fewer pre-existing MDRP-covered molecules in the 2019 period, show the largest estimated effects. The within-class pattern is broadly consistent with differential MDRP coverage as a moderating factor on the magnitude of list price adjustment following the exemption.

6. Discussion

The TRAIN VAT exemption produced no reduction in the prices Filipino patients pay for exempt drugs. Volume-weighted estimates are uniformly negative and insignificant across all license types—originator (-0.002, n.s.), branded non-originator (-0.005, n.s.), unbranded non-originator (-0.026, n.s.)—implying zero pass-through at the margin of actual consumption. The fiscal cost of the exemption accrued entirely to producers as margin expansion on low-volume niche products, generating no consumer surplus at scale.

The null volume-weighted result is explained by the structure of the Philippine pharmaceutical market. High-volume products, which dominate patient spending, are subject to MDRP price ceilings and hospital formulary contracts that prevent list price adjustment regardless of VAT treatment. The positive unweighted estimate ($\hat{\beta} = 0.023^{***}$) reflects list price increases on the unregulated, low-volume tail of the distribution—products with inelastic demand where manufacturers extracted the tax relief as margin without facing competitive or regulatory discipline.

Consistent with standard incidence theory (Weyl and Fabinger, 2013), license type mediates pass-through in the direction predicted by competitive structure. Originator drugs, facing inelastic brand-loyal demand, show the largest positive list price response ($\hat{\beta}_O = 0.045^{***}$), while branded non-originators show a significantly smaller response ($\hat{\beta}_B = 0.013^{**}$), and unbranded generics show no significant adjustment ($\hat{\beta}_U = 0.002$, n.s.). Wald tests confirm a sharp gradient: the originator coefficient is statistically different from both branded non-originator ($p < 0.001$) and unbranded non-originator ($p < 0.001$), while branded and unbranded coefficients are indistinguishable ($p = 0.364$). This originator gradient confirms that brand equity allows manufacturers to absorb the tax relief as margin expansion, and that price competition among generics limits such extraction. The volume-weighted null is uniform across all three tiers: the originator gradient is concentrated in low-volume niche products where MDRP ceilings do not bind.

The therapeutic class decomposition shows positive list price responses across all three treated classes: hypertension ($\hat{\beta} = 0.019^{***}$), high cholesterol ($\hat{\beta} = 0.020^{*}$), and diabetes ($\hat{\beta} = 0.038^{***}$; descriptive due to pre-trend rejection). The within-class pattern is broadly consistent with differential MDRP coverage—diabetes and high cholesterol show larger effects, consistent with fewer regulated molecules in those classes.

The welfare cost of zero pass-through is compounded by a concurrent quantity decline. A parallel difference-in-differences on log standard dosage units sold—identical specification to the price DiD—yields $\hat{\beta}_{qty} = -0.129^{***}$ overall (Appendix Table A6). The channel decomposition is stark: retail quantities fell sharply ($\hat{\beta}_{retail} = -0.208^{***}$, SE = 0.063), while hospital quantities showed no significant change ($\hat{\beta}_{hospital} = -0.015$, n.s.), consistent with hospital formulary procurement decoupling from manufacturer list prices. The dual finding—list prices rose and retail quantities fell—constitutes a double welfare loss for Filipino patients purchasing drugs outside institutional channels.

A rough back-of-envelope calculation illustrates the welfare stakes. Philippine households spent an average of PHP 5,158 on drugs out of pocket in 2012 (Ulep and dela Cruz, 2013); with hypertension, diabetes, and high cholesterol collectively accounting for approximately 45 percent of chronic disease drug expenditure, the annual household outlay on VAT-exempt drugs was of the order of PHP 2,321. Full pass-through of a net fiscal saving of 7–9 percent—the range consistent with partial input-VAT non-recoverability under the exemption design—would have yielded a household saving of approximately PHP 163–209 per year. Instead, the observed positive price effect of +2.3 percent implies that households paid roughly PHP 53 more per year than they would have absent the exemption. The gap between the saving patients should have received and the additional cost they incurred is therefore on the order of PHP 216–262 per household per year, or approximately 4–5 percent of baseline out-of-pocket drug spending. The retail quantity decline amplifies this: a fall of 20.8 percent in retail volumes suggests that some patients reduced consumption or substituted away from listed products, incurring health costs that do not appear in any price index. These figures are necessarily approximate. They rest on list prices recovered from IQVIA data rather than transaction prices, on a household baseline that does not account for changes in disease prevalence or product mix, and on a constant spending share that may shift across income groups. They should be read as order-of-magnitude illustrations of the direction of the welfare impact; the true welfare effect requires household-level survey data linking drug purchase decisions to income and health outcomes.

Lambojon et al. (2020) document in a cross-sectional outlet survey that originator brand prices remained unaffordable in August 2019, eight months after the exemption took effect, and that availability of lowest-priced generics reached only 25 percent in the public sector. Their Discussion section attributes these conditions to “lowered prices” from the

VAT exemption—an assumption our DID estimates directly refute. Prices did not fall relative to what they would have been absent the exemption. Eight months after the exemption took effect—the exact window captured by the Lambojon et al. August 2019 survey ($k = 2$, 2019Q3)—our event study places the treated–control price gap at +1.5 percent ($\hat{\beta}_{k=2} = 0.015$**, SE = 0.006, $p = 0.020$), rising to +2.3 percent by $k = 3$ (2019Q4). The access problem Lambojon et al. (2020) document is therefore not a residual affordability gap that persists despite effective pass-through: it reflects an active price increase. The event study corroborates this directly: at $k = 2$ (2019Q3, the quarter of their outlet survey), the estimated price change is $\hat{\beta}_{k=2} = +0.015^{**}$ (SE = 0.006), implying list prices had already risen approximately 1.5 percent above the pre-exemption baseline by the time their data were collected, reaching the full 2.3 percent by $k = 3$ (2019Q4). Their finding is fully consistent with ours.

These findings carry a direct policy implication. VAT exemptions on pharmaceuticals are an ineffective instrument for reducing patient drug costs in markets where procurement regulation already binds for high-volume products. The fiscal cost falls on the public revenue base while generating no consumer surplus at scale. The quantity response sharpens this welfare assessment. A parallel difference-in-differences on log standard dosage units sold (Appendix Table A6) finds that retail volumes of treated drugs fell approximately 20 percent following the exemption ($\hat{\beta}_{\text{retail}} = -0.208^{***}$, SE = 0.063; pre-trend $F(8)$: $p = 0.203$), while hospital volumes show no significant change ($\hat{\beta}_{\text{hosp}} = -0.015$, n.s.). The channel asymmetry is consistent with formulary procurement insulating hospital acquisition from list-price movements, while retail patients responded to higher prices by reducing purchases or substituting toward non-listed generics. The TRAIN exemption therefore produced a double welfare loss in the retail channel: list prices rose and volumes fell. Future work using transaction-level data, rather than the list prices recovered from IQVIA sales data, would allow direct verification of whether formulary contracts and MDRP ceilings bind at the point of patient purchase.

REFERENCES

Abadie, Alberto, Alexis Diamond, and Jens Hainmueller, "Synthetic Control Methods for Comparative Case Studies: Estimating the Effect of California's Tobacco Control Program," *Journal of the American Statistical Association*, 2010, *105* (490), 493–505.

______, ______, and ______, "`Synth`: An `R` Package for Synthetic Control Methods in Comparative Case Studies," *Journal of Statistical Software*, 2011, *42* (13), 1–17.

______, ______, and ______, "Comparative Politics and the Synthetic Control Method," *American Journal of Political Science*, 2015, *59* (2), 495–510.

Batangan, Dennis B., C. Echavez, A.A. Santiago, A.C. de la Cruz, and E. Santos, "The Prices People Have to Pay for Medicines in the Philippines," Technical Report, Ateneo de Manila University Institute of Philippine Culture, Quezon City, Philippines 2005.

Benzarti, Youssef and Dorian Carloni, "Who Really Benefits from Consumption Tax Cuts? Evidence from a Large VAT Reform in France," *American Economic Journal: Applied Economics*, 2019, *11* (1), 44–76.

Bureau of Internal Revenue, "Revenue Regulations No. 25-2018: Implementing the Value-Added Tax Exemption on the Sale of Drugs and Medicines Prescribed for Diabetes, High Cholesterol and Hypertension under Republic Act No. 10963," December 2018. Republic of the Philippines.

Callaway, Brantly and Pedro H. C. Sant'Anna, "Difference-in-Differences with Multiple Time Periods," *Journal of Econometrics*, 2021, *225* (2), 200–230.

Department of Finance and Department of Health and Bureau of Internal Revenue and Food and Drug Administration, "Joint Administrative Order No. 2-2018: Guidelines on the Implementation of Value-Added Tax Exemption on Drugs and Medicines for Diabetes, High Cholesterol and Hypertension," 2018. Republic of the Philippines.

Embassy of Italy in Manila, "Philippines Pharmaceuticals Sector Guide," Technical Report, Embassy of Italy in Manila 2023. Prepared for the Embassy of Italy in Manila.

Food and Drug Administration, "FDA Advisory No. 2019-122: List of VAT-Exempt Diabetes, High Cholesterol and Hypertension Drugs," May 2019. Republic of the Philippines. Available at https://www.fda.gov.ph/fda-advisory-no-2019-122.

Goodman-Bacon, Andrew, "Difference-in-Differences with Variation in Treatment Timing," *Journal of Econometrics*, 2021, *225* (2), 254–277.

Lambojon, Krizzia, Jie Chang, Amna Saeed, Khezar Hayat, Pengchao Li, Minghuan Jiang, Naveel Atif, Gebrehaweria Kassa Desalegn, Faiz Ullah Khan, and Yu Fang, "Prices, Availability and Affordability of Medicines with Value-Added Tax Exemption: A Cross-Sectional Survey in the Philippines," *International Journal of Environmental Research and Public Health*, 2020, *17* (14), 5242.

Magno, Cielo and Ricardo Rafael S. Guzman, "Pharmaceutical Competition in the Philippines," Discussion Paper 2022-01, Philippine Competition Commission, Pasay, Philippines 2022.

Philippine Statistics Authority, "Consumer Price Index, All Income Households, Philippines (Base Year 2018 = 100)," Table 2M4ACP22. Retrieved from https://psa.gov.ph/price-indices/cpi-ir 2026. Monthly series, January 2018–March 2026.

Republic of the Philippines, "An Act Providing for Cheaper Medicines, Amending for the Purpose Sections 50, 72, 76 and 92 of Republic Act No. 8293, Otherwise Known as the "Intellectual Property Code of the Philippines"," Republic Act No. 9502 2008. Cheaper Medicines Act of 2008.

———, "Tax Reform for Acceleration and Inclusion," Republic Act No. 10963 2017. TRAIN Law; Section 34 amends NIRC Section 109(AA).

Reyes, Celia M. and Aubrey D. Tabuga, "Overview of the Philippine Pharmaceutical Market," Discussion Paper, Philippine Institute for Development Studies, Quezon City, Philippines 2018.

Sarol, Jesus N., "Effect of Government Mediated Access Pricing on Prices of Targeted Drugs in the Philippines," *Journal of Asian Scientific Research*, 2014, *4* (9), 473–489.

Sun, Liyang and Sarah Abraham, "Estimating Dynamic Treatment Effects in Event Studies with Heterogeneous Treatment Effect," *Journal of Econometrics*, 2021, *225* (2), 175–199.

Ulep, Valerie Gilbert T. and Nina Ashley O. dela Cruz, "Analysis of Out-of-Pocket Expenditures in the Philippines," *Philippine Journal of Development*, 2013, *40* (1–2), 93–123.

Weyl, E. Glen and Michal Fabinger, "Pass-Through as an Economic Tool: Principles of Incidence under Imperfect Competition," *Journal of Political Economy*, 2013, *121* (3), 528–583.

World Bank, "Inflation, Consumer Prices (Annual %), Philippines," World Development

Indicators. Retrieved from https://data.worldbank.org/indicator/FP.CPI.TOTL.ZG 2026.

Appendix A. Anticipation and Robustness Tables

Table A1: Anticipation Test:
Pre-Announcement Price Response

	Preferred F(8) spec
$\hat{\beta}$ (Treated × Post-announce)	0.0085*
	(0.0048)
Pre-announce F p	0.203
Pre-announce F d.f.	$F(8)$
Observations	21 229
No. products	1717
Product FE	Yes
Quarter FE	Yes
Sector FE	Yes
Product trends	Yes
Adj. R^2	0.9993

Dependent variable: log real drug price (2012 PHP). Sample restricted to 2016Q4–2018Q4. Post-announce = 1 for 2018Q1–2018Q4 (TRAIN Act signed December 2017). Pre-period = 2016Q4–2017Q4. Wave 2 controls, product trends. Clustered SEs at product level.

Marginal anticipatory repricing ($p = 0.075$) suggests pre-implementation adjustment attenuates rather than inflates the main result: excluding 2018 would raise $\hat{\beta}$ from 0.023 to approximately 0.036.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A2: Robustness:
Pre-COVID Window Only (2016Q4–2019Q4)

	(1) Wave 2	(2) Wave 2†	(3) J01
$\hat{\beta}$ (Treated × Post)	0.0307***	0.0117***	0.0083**
	(0.0055)	(0.0044)	(0.0042)
Pre-trend F p	—	0.104	—
Observations	30 402	30 402	41 789
No. products	1823	1823	2815
Product FE	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes
Sector FE	Yes	Yes	Yes
Product trends	No	Yes	Yes
Adj. R^2	0.9988	0.9993	0.9982

Dependent variable: log real drug price (2012 PHP). Sample restricted to 2016Q4–2019Q4 to exclude COVID-19 disruptions. Clustered SEs at product level.

Stability of $\hat{\beta}$ relative to Table 1 indicates COVID-period price dynamics do not drive the main result.

† Preferred specification (Wave 2, product trends).

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A3: Identification Progression: V-Shape in $\hat{\beta}$ and Pre-Trend F

	(1) All ctrl 2009+	(2) Wave 2 2015+ no trends	(3) Wave 2 2015+ trends	(4) Wave 2† 2016Q4+	(5) Wave 2 2017Q4+
$\hat{\beta}$ (Treated × Post)	0.0293***	0.0263**	0.0168**	0.0228***	0.0295***
	(0.0077)	(0.0104)	(0.0076)	(0.0052)	(0.0072)
Pre-trend F p	<0.001	0.028	0.055	0.081	0.281
Pre-trend F d.f.	$F(36)$	$F(15)$	$F(15)$	$F(8)$	$F(4)$
Pre-period start	2009Q3	2015Q1	2015Q1	2016Q4	2017Q4
Product trends	No	No	Yes	Yes	Yes
Observations	181 707	60 538	60 538	44 339	34 822
No. products	5357	2141	2141	1953	1832
Adj. R^2	0.9961	0.9971	0.9985	0.9988	0.9989

Dependent variable: log real drug price (2012 PHP). Columns (1)–(2) use the 2009+ or 2015Q1+ pre-period, no product trends. Column (3) adds product trends; column (4) shortens to 2016Q4+ (F(8))†; column (5) further shortens to 2017Q4+ (F(4)). Clustered SEs at product level.

The V-shape in $\hat{\beta}$ reflects identification quality: (1)–(2) are inflated by poor parallel trends; (3)–(4) tighten identification; (5) passes the F-test mechanically due to only 4 pre-period dummies.

† Preferred specification.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

A.1. Identification Discipline Progression.

Table A4: Volume-Weighted Specification: Pre-Trend Diagnostic

	(1) UW 2015Q1+	(2) UW 2016Q4+†	(3) VW 2015Q1+	(4) VW 2016Q4+
$\hat{\beta}$ (Treated × Post)	0.0168**	0.0228***	−0.0080	−0.0427
	(0.0076)	(0.0052)	(0.0158)	(0.0398)
Pre-trend F p^{b}	0.055	0.081	<0.001	<0.001
Pre-trend F d.f.	$F(15)$	$F(8)$	$F(15)$	$F(8)$
Volume-weighted	No	No	Yes	Yes
Observations	60 538	44 339	60 538	44 339
No. products	2141	1953	2141	1953
Adj. R^2	0.9985	0.9988	0.9968	0.9970

Dependent variable: log real drug price (2012 PHP). All columns include product trends. Cols (1)–(2): unweighted (equal product weight). Cols (3)–(4): weighted by standard dosage units sold. Clustered SEs at product level.

Volume-weighted specifications fail the pre-trend F-test across both window lengths, indicating that the weighted composition of products changes differentially between treatment and control groups before the reform. High-volume products are disproportionately subject to MDRP price ceilings, mechanically suppressing post-reform price responses in the weighted sample.

† Preferred specification.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

A.2. Volume-Weighted Pre-Trend Diagnostic.

A.3. CREATE Act Placebo and Quantity Tests.

Table A5: Placebo Test: CREATE Act (RA 11534) and Wave 2 Control Prices

	Wave 2 vs. J01
$\hat{\beta}$ (Wave 2 × Post-CREATE)	−0.0010
	(0.0064)
Pre-trend $F(16)$ p^{c}	0.030
Wave-2-only ITS collinear with quarter FE[b]	Yes
Observations	43 539
No. products	2302
Product FE	Yes
Quarter FE	Yes
Sector FE	Yes
Product trends	Yes

Dependent variable: log real drug price (2012 PHP). Wave 2 control products (L01, N05, N06, J04) and J01 (antibiotics) as control group, 2016Q4–2021Q3. `post_create` = 1 from 2021Q1 (RA 11534 effective date). Clustered SEs at product level.

The within-group ITS is not separately identified: `post_create` is perfectly collinear with the quarter fixed effects because all Wave 2 products receive the CREATE exemption at the same calendar quarter. This collinearity is the intended diagnostic: the quarter FEs absorb the 2021Q1 level shift, confirming it reflects a common time trend (product attrition/selection) rather than a Wave-2-specific price response. Coefficient shown is from the DiD specification with J01 as comparison group; see text for interpretation.

Pre-trend F-test from the DiD event-study specification with J01 control; p < 0.05 indicates J01 prices also shifted around 2021, invalidating the DiD comparison and further supporting the collinearity diagnosis.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A6: Quantity Response to VAT Exemption (F(8) Preferred Specification)

	(1) Full	(2) Retail	(3) Hospital
$\hat{\beta}$ (Treated × Post)	−0.1287***	−0.2078***	−0.0153
	(0.0436)	(0.0629)	(0.0505)
Pre-trend $F(8)$ p^{b}	0.203	—	—
Observations	44 339	18 460	25 733
No. products	1953	1423	1819
Product FE	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes
Sector FE	Yes	Yes	Yes
Product trends	Yes	Yes	Yes
Adj. R^2	0.8330	0.8644	0.9117

Dependent variable: log(standard dosage units sold). Preferred specification: Wave 2 controls, 2016Q4–2021Q3, product-specific linear trends. Clustered standard errors at product level in parentheses.

Pre-trend $F(8)$ p-value from the full-sample quantity event study. $p = 0.203$ supports parallel pre-trends in quantities over the pre-exemption window.

Columns (2)–(3) estimate separate regressions on retail and hospital sub-samples to allow sector-specific trends; they do not use a triple interaction with sector.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A7: Placebo-in-Time:
(Fake Treatment Assigned to 2017Q1)

	Fake event 2017Q1
$\hat{\beta}$ (Treated × Fake Post)	0.0062
	(0.0043)
Real pre-trend $F(8)$ p (for comparison)	0.081
Observations	21 229
No. products	1717
Product FE	Yes
Quarter FE	Yes
Sector FE	Yes
Product trends	No

Dependent variable: log real drug price (2012 PHP). Fake treatment assigned to 2017Q1, two years before the actual VAT exemption. Panel restricted to the pre-period (2016Q4–2018Q4). Specification uses product and quarter fixed effects without product-specific linear trends (trends absorb all variation in the short pre-period window). Clustered SEs at product level.

The absence of a significant coefficient ($p = 0.146$) confirms that no pre-existing divergence between treated and control classes would generate a spurious result at an arbitrary pre-treatment date. The actual treatment date (2019Q1) is the only date that produces a significant positive estimate.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

A.4. Placebo-in-Time Test.

Appendix B. Synthetic Control: Data Construction and Results

The synthetic control uses the IQVIA MIDAS World Review Pack for Indonesia, Malaysia, Singapore, and Thailand. For each therapeutic class, we construct a quarterly log mean price index by averaging `ln_price` (log nominal USD per standard dosage unit) across all products within the class for each country–quarter cell. Products are matched at the ATC2 level: all products with ATC2 codes C07, C08, or C09 enter the hypertension index; A10 enters the diabetes index; C10 enters the cholesterol index. No further product-level harmonisation is applied across countries, so the index reflects each country's available product mix within the class rather than a matched set of identical molecules. The outcome variable is therefore the within-class mean log price, which is comparable across countries to the extent that the therapeutic class composition is similar.

Pre-treatment matching uses four predictors: mean log price over 2009Q4–2012Q3, mean log price over 2012Q4–2015Q4, mean log price over 2016Q1–2018Q4, and the 2018Q4 level. No external macroeconomic predictors (GDP, exchange rates, health expenditure) are available at the country–quarter level for this sample; matching on outcome lags only is standard when covariate data are unavailable (Abadie et al., 2010) Synthetic weights are estimated by minimising the pre-treatment root mean squared prediction error (RMSPE) via the `Synth` package in `R` (Abadie et al., 2011). Placebo inference assigns the treatment sequentially to each of the four donor countries, with the Philippines rejoining the donor pool; the permutation p-value is the share of units (including the Philippines) whose post/pre RMSPE ratio equals or exceeds that of the Philippines.

B.1. Donor Weights and Fit Statistics.

Table B-8: Synthetic Control: Donor Weights and Fit Statistics

	Hypertension	Diabetes	Cholesterol
Panel A: Donor weights			
Indonesia	1.000	—	1.000
Malaysia	—	0.112	—
Singapore	—	0.888	—
Thailand	—	—	—
Panel B: Fit statistics			
Pre-period RMSPE	0.4088	0.1197	0.2572
Post-period RMSPE	0.4650	0.1048	0.3942
Ratio (post/pre)	1.14	0.88	1.53
Avg. post gap	0.4642	0.0344	0.3885

Synthetic control estimated via the `Synth` package (Abadie et al., 2011)). Donor pool: Indonesia, Malaysia, Singapore, Thailand. Outcome-only predictors: mean log price index in three pre-treatment sub-periods plus 2018Q4 level (**?**).

RMSPE = root mean squared prediction error. Pre-period: 2009Q4–2018Q4 (37 quarters). Post-period: 2019Q1–2021Q3 (11 quarters). Ratio > 1 indicates worse post-treatment fit, consistent with a treatment effect.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (permutation p-values from in-place placebo inference reported in text).